

Exploring and Coordinating Optimal Team Sizes with Thymio Robots Swarm

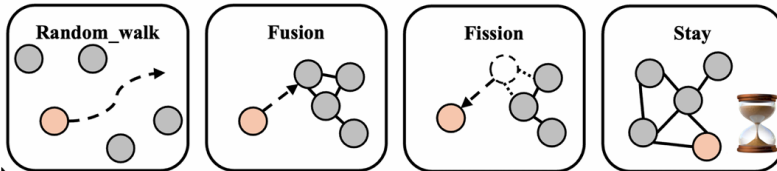
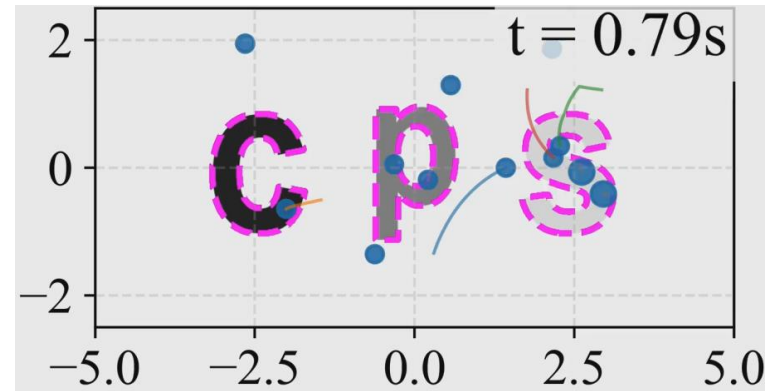
Presentation by Brendan Bilodeau – Robotics Engineering

The goal of my internship :

Thymio robot and
Fission-Fusion based controller



CPS experiment



Sensors



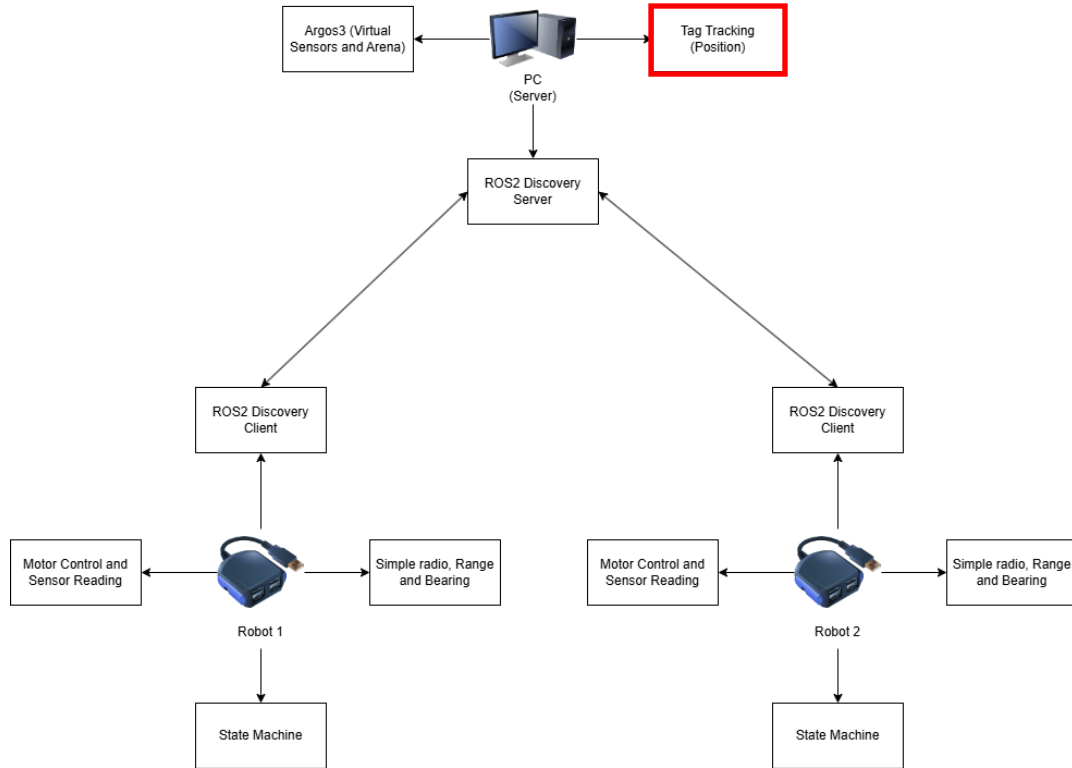
Ground sensors
(Greyscale readings)



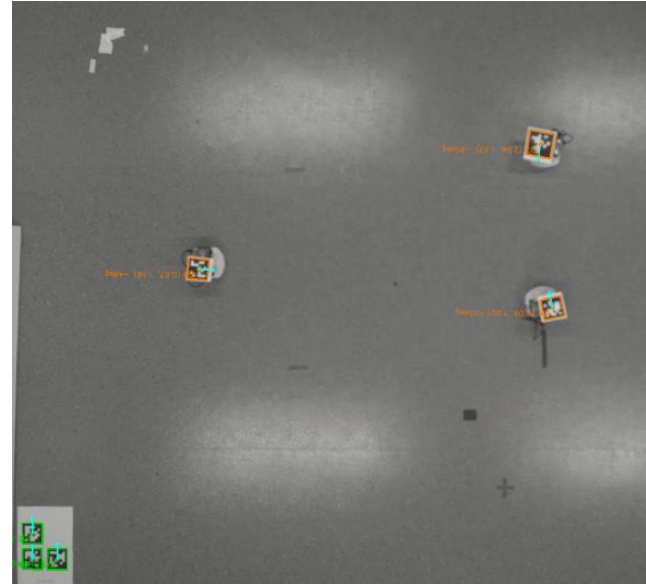
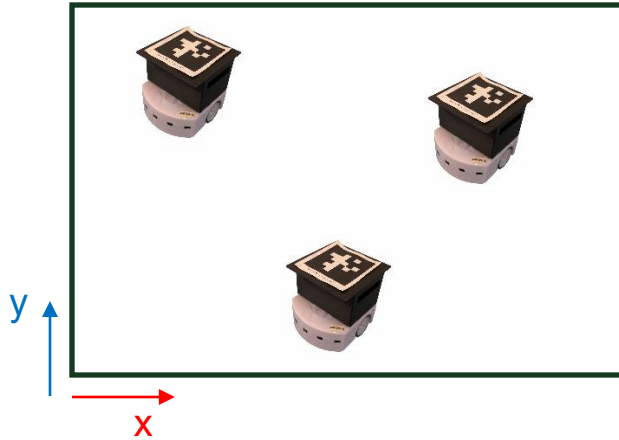
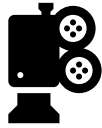
Simple Radio
Range and bearing
Perception Sensor



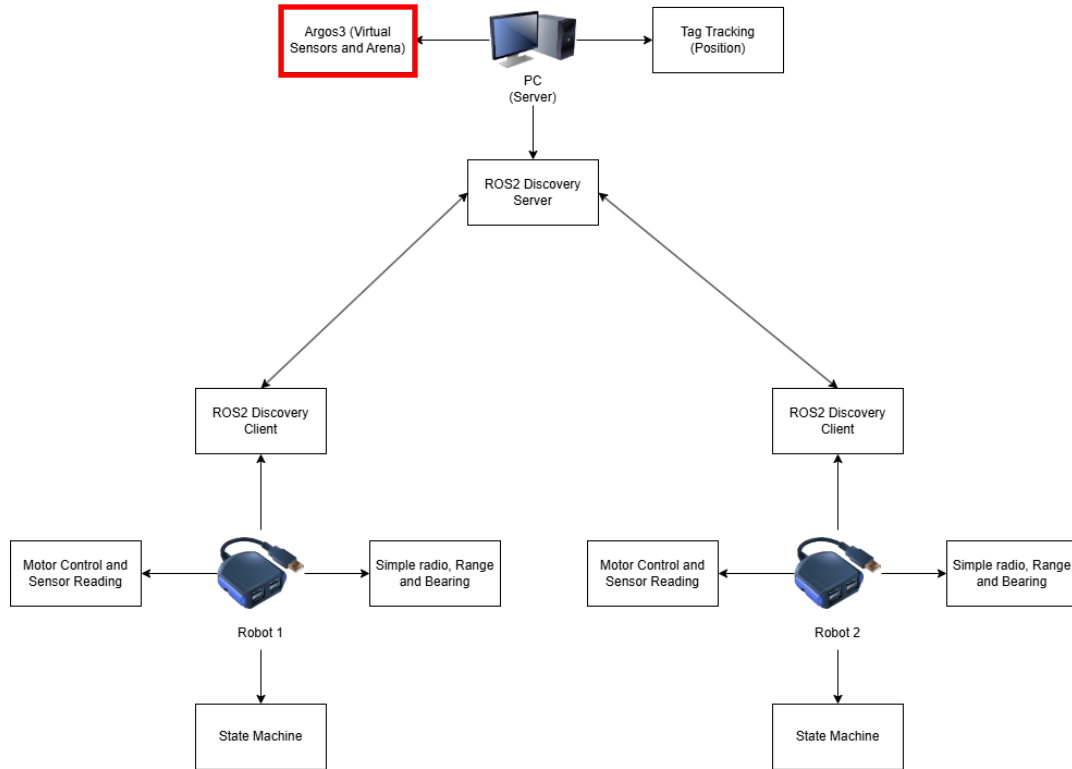
Global Architecture



Tracking

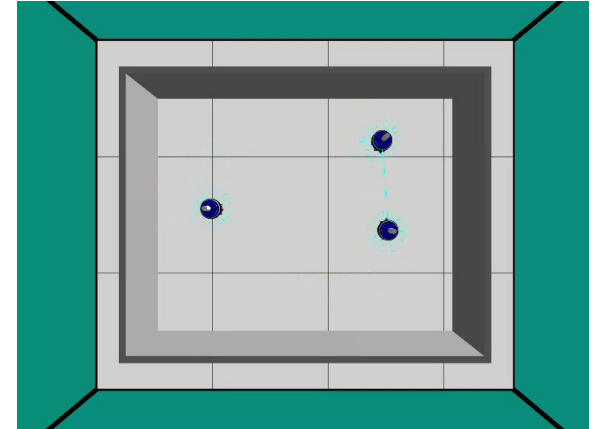
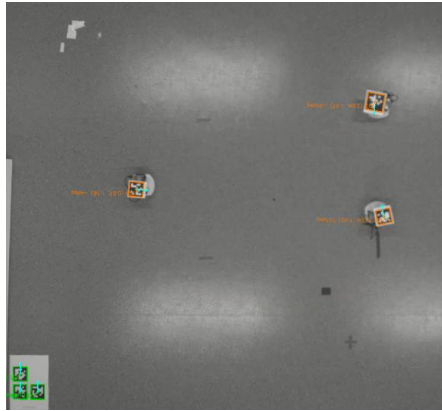


Global Architecture

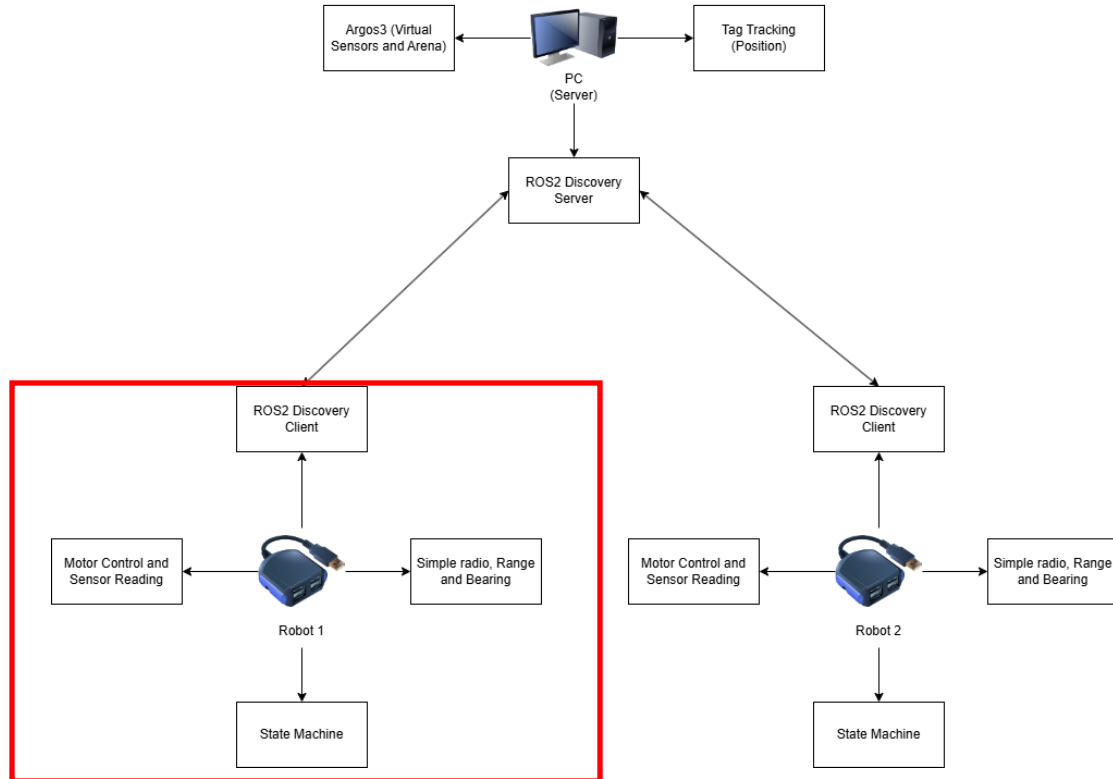


ARGoS3

- Virtual Boundaries
- Virtual Sensors
- Simple Radio
- Range and bearing

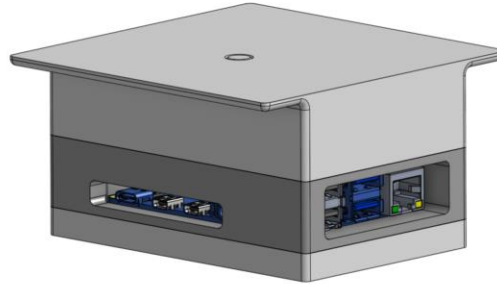
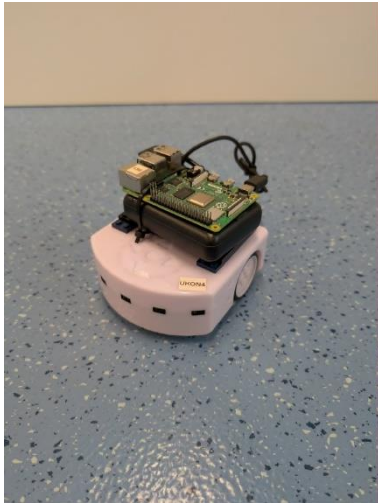


Global Architecture



3D Modeling

Before

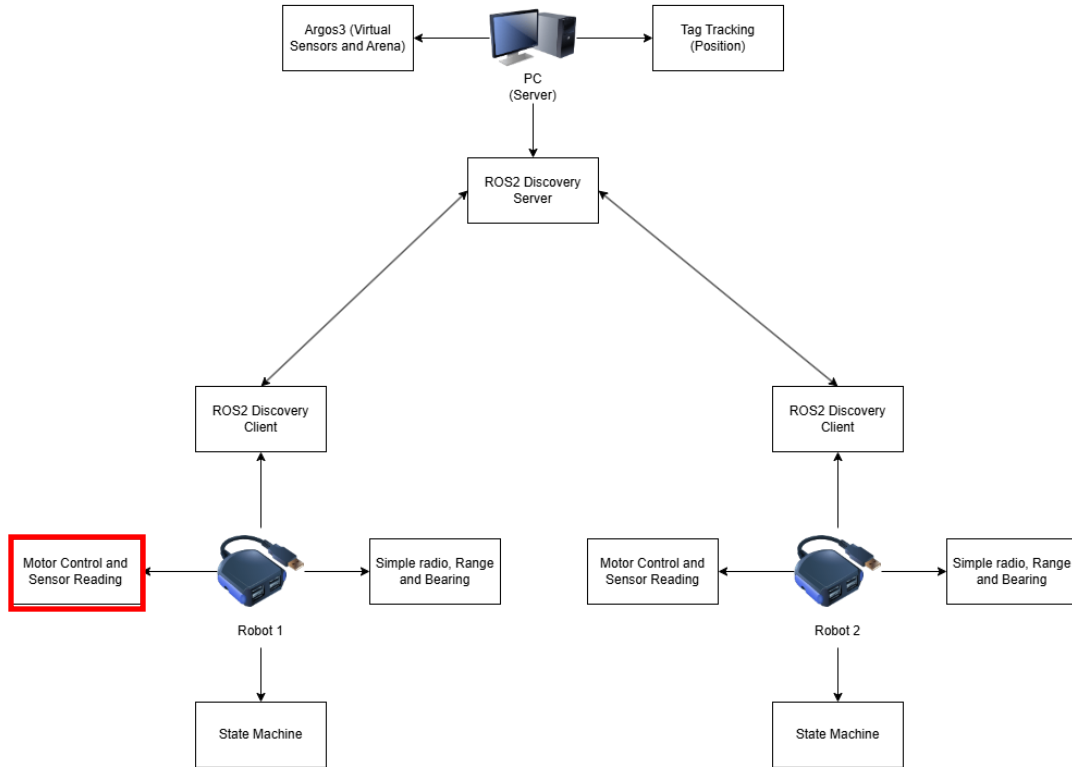


After



- More protection
- Helps with tracking
- Better style

Global Architecture

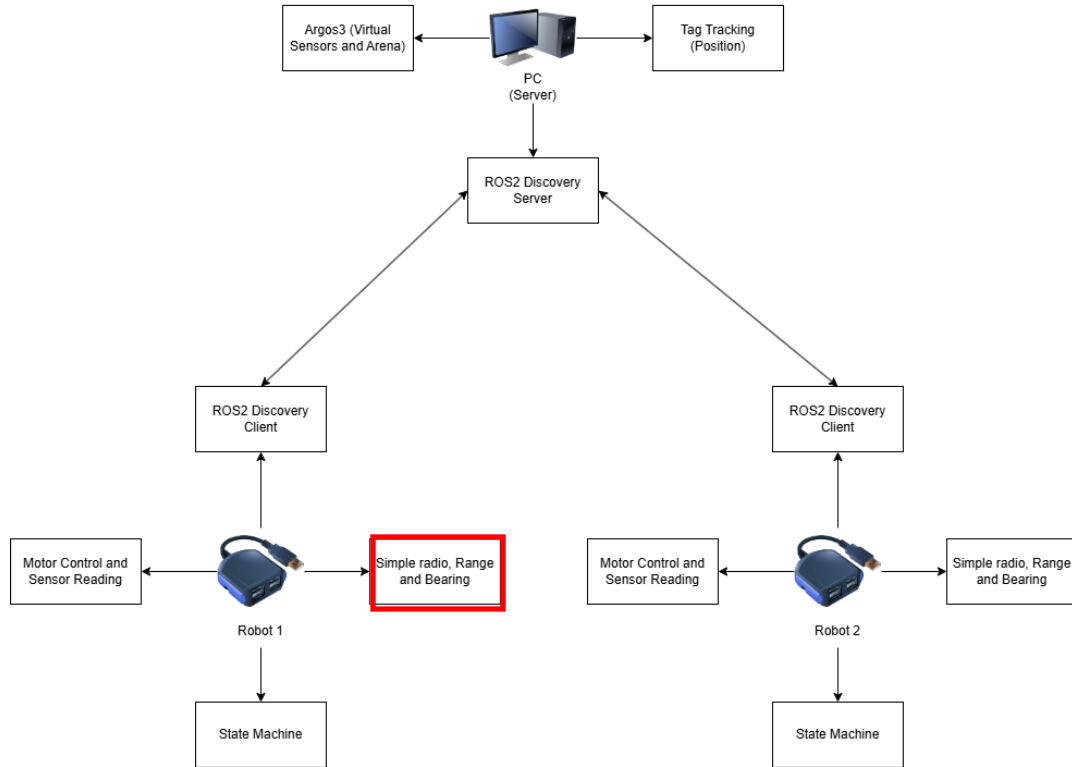


Motor Controller

$$\overrightarrow{V_{final}} = w_1 \overrightarrow{v_1} + w_2 \overrightarrow{v_2} + w_3 \overrightarrow{v_3} + \dots + w_n \overrightarrow{v_n}$$

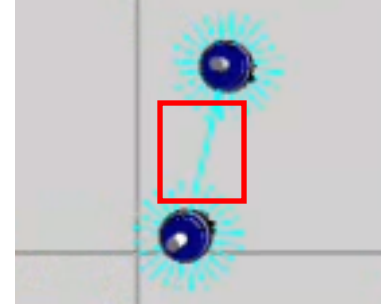
- Wall avoidance
- Robot avoidance
- Letter following
- Random Walk
- Fusion
- Fission

Global Architecture

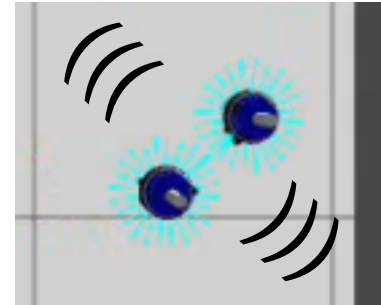


Range and Bearing – Simple Radio

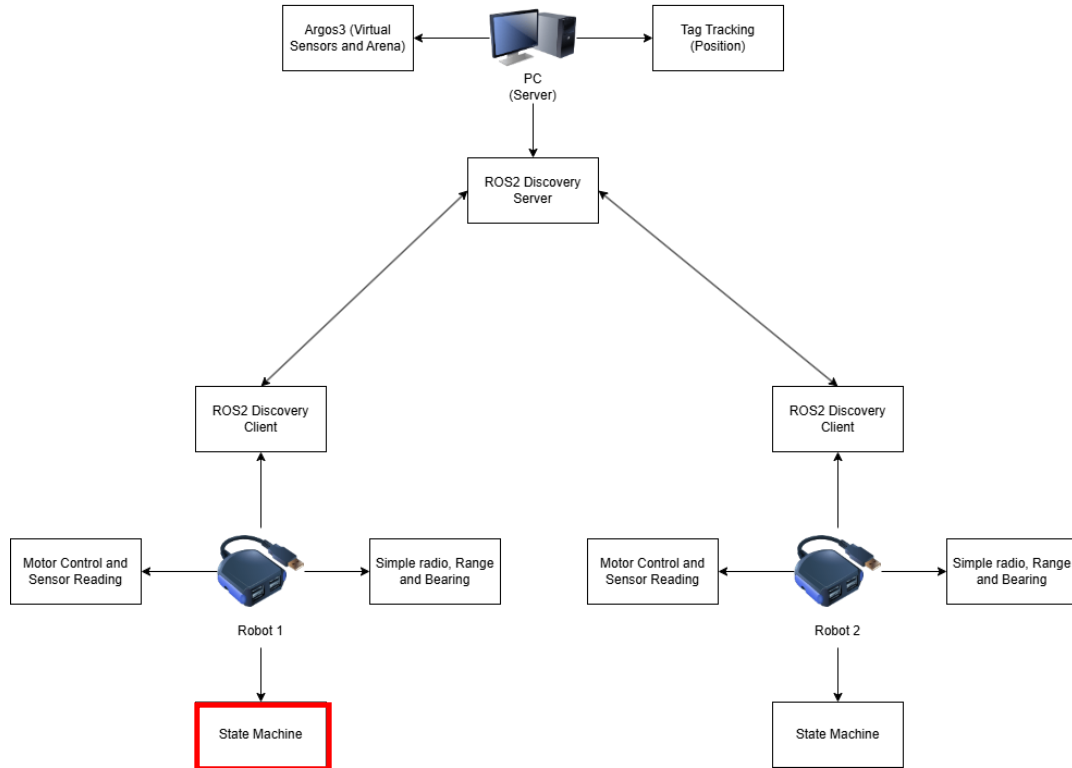
- Range and Bearing :
- Position Sharing
 - Current state



- Simple Radio :
- Estimated group size in Stay state

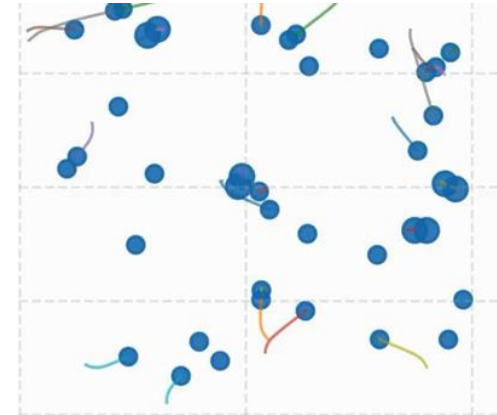


Global Architecture

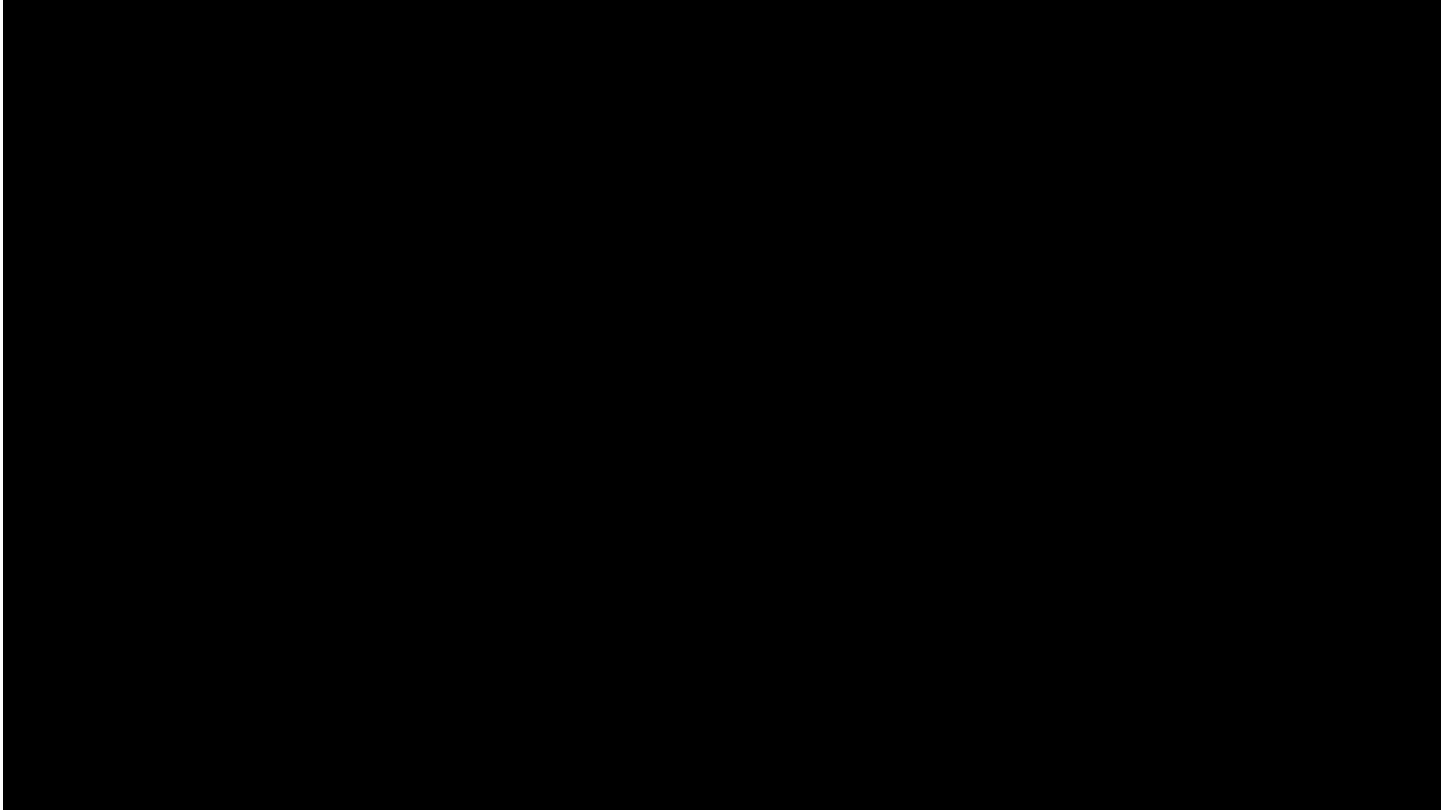


Testing phase - Coalition

Current State	Random Walk	Fusion	Stay	Fission
Random Walk	-	Find target	-	-
Fusion	No target	-	Close to target	Group size estimation > Group size wanted
Stay	-	-	-	Time exceeded
Fission	Far from other robots	-	-	-

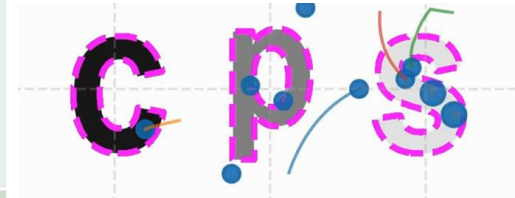


Coalition – Example with desired group size of 2

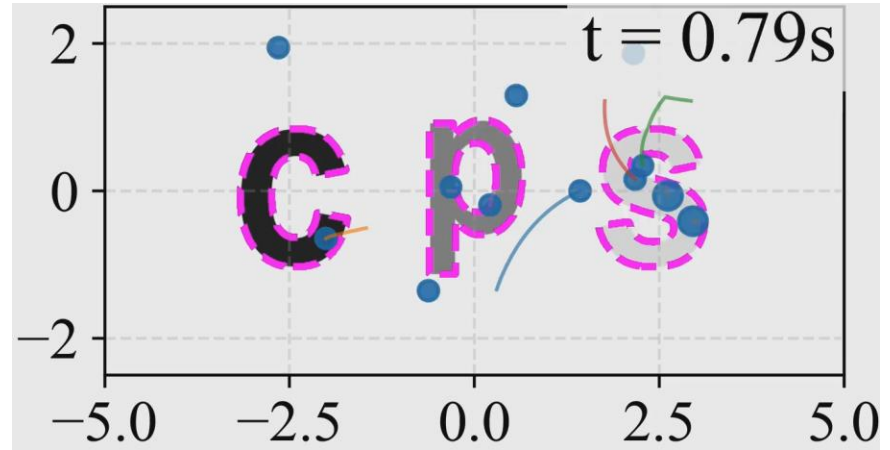
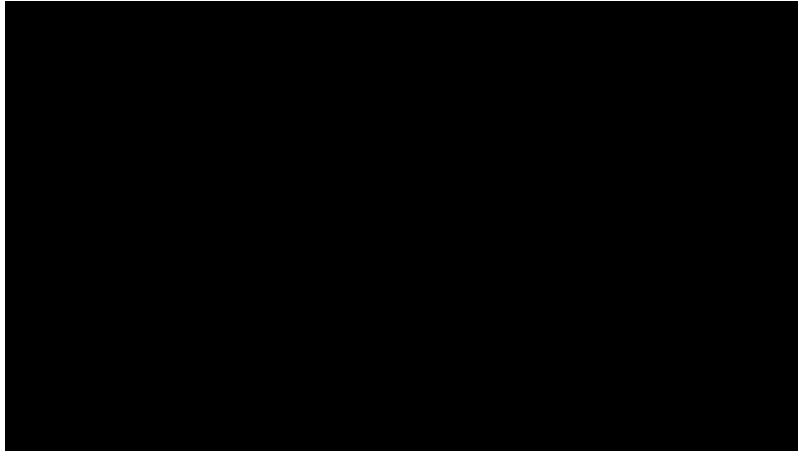


Allocation – CPS experiment

Current State	Random Walk	Fusion	Stay	Fission
Random Walk	-	Find target in Stay	Find letter	-
Fusion	No target	-	Find letter	Group size estimation > Group size wanted
Stay	-	-	-	Time exceeded or Group size estimation > Group size wanted
Fission	Far from other robots and not on a letter	-	-	-



CPS experiment test



Next Steps

- Do the experiments with 12 robots
- Bigger and better letters
- Compare real data with simulation data
- Document the code for easy takeover

Thank you!

